

Q3a) List and Explain the Services Offered by Amazon Web Services

AWS (Amazon Web Services) is a cloud computing platform that provides various on-demand services such as computing power, storage, databases, networking, and security over the internet. **Major Services Offered by AWS -**

1. EC2 (Elastic Compute Cloud)

- Provides virtual servers in the cloud.
- Users can launch and manage virtual machines called instances.
- **Features** - Scalable computing, Different instance types, Pay-as-you-use

2. S3 (Simple Storage Service)

- Object storage service used to store files and backups.
- **Features** - High durability, Secure storage, Easy accessibility

3. RDS (Relational Database Service)

- Managed relational database service.
- **Supports Databases** - MySQL, PostgreSQL, Oracle, SQL Server

4. Lambda

- Serverless computing service.
- Runs code without managing servers.
- **Advantages** - Automatic scaling, Cost efficient

5. VPC (Virtual Private Cloud)

- Creates isolated private networks in AWS cloud.
- **Features** - Improved security, Network control

6. CloudFront

- Content Delivery Network (CDN).
- Delivers web content with low latency.

7. IAM (Identity and Access Management)

- Manages users and permissions securely.
- **Features** - Authentication, Access control

8. Elastic Load Balancer (ELB)

- Distributes incoming traffic across multiple servers.

9. DynamoDB

- Fully managed NoSQL database service.
- **Features** - Fast performance, Automatic scaling

10. CloudWatch

- Monitoring service for AWS resources and applications.

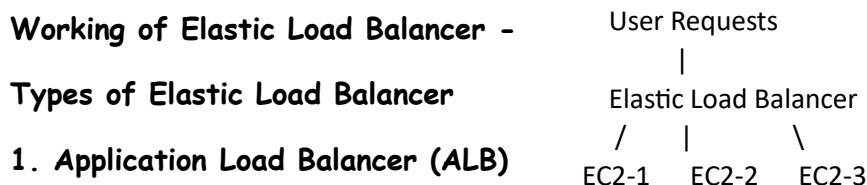
Advantages of AWS Services - Scalability, Reliability, Security, Cost effectiveness, Global accessibility.

AWS provides a wide range of cloud services that help organizations build scalable, reliable, and secure applications efficiently.

Q3b) Explain the AWS Elastic Load Balancer and its Types with Advantages

Elastic Load Balancer (ELB) is an AWS service that automatically distributes incoming application traffic across multiple targets such as EC2 instances, containers, or IP addresses.

It improves: Availability, Scalability, Fault tolerance



- Operates at Layer 7 (Application Layer).
- Routes traffic based on: URL, Hostname, HTTP headers
- **Used For** Web applications, Microservices

2. Network Load Balancer (NLB)

- Operates at Layer 4 (Transport Layer).
- Handles TCP and UDP traffic.
- **Features** - Very high performance, Low latency

3. Classic Load Balancer (CLB)

- Older generation load balancer.
- Supports both Layer 4 and Layer 7.
- **Used For** Legacy applications

4. Gateway Load Balancer (GWLB)

- Used for deploying and managing virtual network appliances.
- **Features** - Security inspection, Traffic monitoring

Advantages of Elastic Load Balancer

- 1. High Availability** - Distributes traffic across multiple servers.
- 2. Fault Tolerance** - Automatically redirects traffic from failed instances.

3. **Scalability** - Automatically adjusts according to traffic load.

4. **Improved Performance** - Reduces server overload.

5. **Security** - Supports SSL/TLS termination.

6. **Health Monitoring** - Continuously checks server health.

AWS Elastic Load Balancer efficiently distributes workloads across multiple servers, ensuring high availability, scalability, and reliability of cloud applications.

Q3c) Draw and Explain Architecture of Amazon Dynamo

Amazon Dynamo is a highly available and scalable distributed NoSQL key-value storage system developed by Amazon.

It is designed for: High availability, Fault tolerance, Scalability

Architecture of Amazon Dynamo -

Components of Dynamo Architecture

1. **Client** - Sends read/write requests to Dynamo.

2. **Coordinator Node**

- Receives client request.
- Coordinates data storage and retrieval.

3. **Distributed Nodes**

- Data is distributed across multiple nodes.
- Each node stores a portion of data.

4. **Replication Mechanism**

- Copies data to multiple nodes.
- Ensures high availability and fault tolerance.

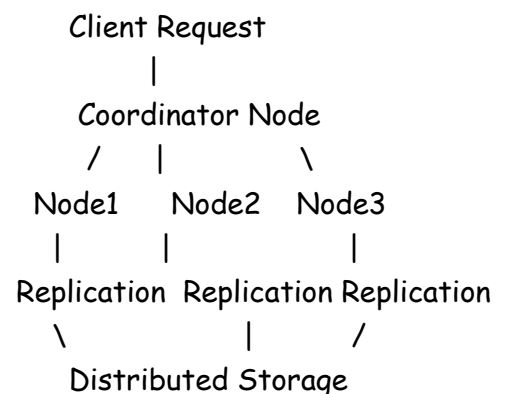
5. **Consistent Hashing**

- Used for data partitioning.
- Efficiently distributes data among nodes.

Key Features of Amazon Dynamo -

1. **Decentralization** - No single point of failure.

2. **Scalability** - Nodes can be added dynamically.



3. Fault Tolerance - System continues working even if some nodes fail.

4. High Availability - Data replicated across multiple servers.

5. Eventual Consistency - Data becomes consistent over time.

Working of Dynamo

1. Client sends request.
2. Coordinator node processes request.
3. Data is partitioned using consistent hashing.
4. Data replicated to multiple nodes.
5. Response sent back to client.

Advantages of Dynamo - High scalability, Reliable distributed storage, Fast data access, Fault tolerant

Amazon Dynamo is a distributed NoSQL database architecture designed to provide scalable, highly available, and fault-tolerant storage for cloud applications.

Q4a) Explain Steps to Configure Server for EC2

Amazon EC2 is a web service provided by AWS that allows users to create and manage virtual servers called instances in the cloud.

Steps to Configure Server for EC2 -

Step 1: Login to AWS Console

- Open AWS Management Console.
- Sign in using AWS account credentials.

Step 2: Open EC2 Dashboard

- Select **EC2** service from AWS services menu.

Step 3: Launch Instance

- Click on **Launch Instance** button.

Step 4: Choose Amazon Machine Image (AMI)

- Select operating system image such as: Ubuntu, Windows, Amazon Linux

AMI contains: OS, Application software, Configuration settings

Step 5: Choose Instance Type

- Select hardware configuration.

- Examples: t2.micro, t2.small
- Defines: CPU, RAM, Network performance

Step 6: Configure Instance Details

Set parameters such as: Number of instances, Network settings, Auto-scaling options

Step 7: Add Storage

- Configure storage volume size and type.

Examples: SSD, Magnetic storage

Step 8: Add Tags

- Assign name and labels to instance for identification.

Example: Name = WebServer

Step 9: Configure Security Group

- Define firewall rules.

Examples: HTTP → Port 80, HTTPS → Port 443, SSH → Port 22

Step 10: Create Key Pair

- Generate/download key pair (.pem file).
- Used for secure login to EC2 instance.

Step 11: Launch Instance

- Review configuration.
- Click Launch.

Step 12: Connect to EC2 Instance

For Linux: `ssh -i key.pem ec2-user@public-ip`

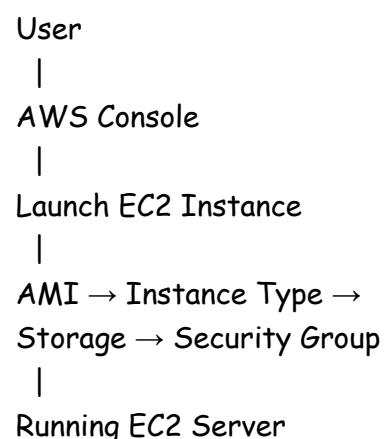
For Windows: Use Remote Desktop Protocol (RDP).

EC2 Configuration Diagram -

Advantages of EC2 - Scalable, Pay-as-you-use, Flexible

Secure, Easy deployment.

EC2 enables users to quickly configure and deploy virtual servers in the cloud with scalable and secure infrastructure.



Q4b) Describe the Storage Services Offered by Amazon Web Services

AWS provides various storage services for storing files, databases, backups, and application data securely and efficiently.

Storage Services Offered by AWS -

1. Amazon S3 (Simple Storage Service) -

- Object storage service used to store files and backups.
- **Features** - High durability, Scalability, Secure storage
- **Applications** - Backup, Media storage, Data archiving

2. Amazon EBS (Elastic Block Store)

- Block-level storage attached to EC2 instances.
- **Features** - Persistent storage, High performance, SSD support
- **Used For** Databases, Virtual machines

3. Amazon EFS (Elastic File System)

- Managed file storage service.
- **Features** - Shared file access, Scalable file system, Multiple EC2 access

4. Amazon Glacier

- Low-cost archival storage service.
- **Features** - Long-term backup, Data archiving, High durability

5. AWS Storage Gateway

- Hybrid storage service connecting on-premises storage to AWS cloud.
- **Features** - Backup integration, Disaster recovery

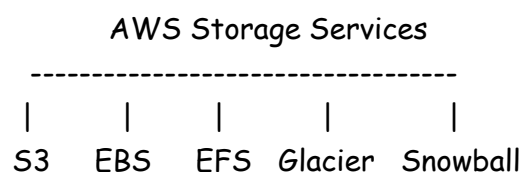
6. AWS Backup

- Centralized backup management service.
- **Features** - Automated backups, Easy recovery

7. Snowball

- Physical device for transferring large amounts of data to AWS.
- **Used For** Offline data migration

Storage Services Diagram -



Advantages of AWS Storage Services - Scalability, High availability, Security, Cost efficiency, Data durability

AWS offers multiple storage services designed for different needs such as object storage, block storage, file storage, archival, and backup solutions.

Q4c) Write a Note on Services Offered by Microsoft

Microsoft provides cloud computing services through **Microsoft Azure** platform. Azure offers services for computing, storage, networking, databases, analytics, and security.

Major Services Offered by Microsoft Azure

1. Azure Virtual Machines

- Provides virtual servers in cloud.
- **Features** - Windows and Linux support, Scalable infrastructure

2. Azure Storage

- Cloud storage service for files, blobs, queues, and disks.
- **Features** - Secure storage, High availability

3. Azure SQL Database

- Managed relational database service.
- **Features** - Automatic backup, Scalability

4. Azure Active Directory (Azure AD)

- Identity and access management service.
- **Features** - Authentication, Single sign-on

5. Azure App Services

- Platform for developing and hosting web applications.

6. Azure Functions

- Serverless computing service.
- **Features** - Event-driven execution, Automatic scaling

7. Azure Networking Services

- **Services Include** - Virtual Network, Load Balancer, VPN Gateway

8. Azure Kubernetes Service (AKS)

- Managed container orchestration service.

9. Azure Backup and Recover

- Backup and disaster recovery services.

10. Azure AI and Analytics Services

- **Services Include** - Machine Learning, Cognitive Services, Data Analytics

Advantages of Microsoft Azure - Scalability, Security, Hybrid cloud support, Global availability, Easy integration with Microsoft products

Microsoft Azure provides a wide range of cloud services for computing, storage, networking, databases, security, AI, and analytics, helping organizations build scalable and reliable cloud applications.

Q3a) Describe the Amazon Database Services

Amazon Web Services provides various managed database services for relational, NoSQL, in-memory, graph, and data warehousing applications.

These services reduce database administration tasks such as: Hardware setup, Backup, Scaling, Maintenance

Amazon Database Services -

1. Amazon RDS (Relational Database Service)

- Managed relational database service.
- **Supported Databases** - MySQL, PostgreSQL, Oracle, MariaDB, SQL Server
- **Features** - Automated backup, Scalability, High availability

2. Amazon DynamoDB

- Fully managed NoSQL database service.
- **Features** - Key-value and document database, High speed performance, Automatic scaling
- **Applications** - Gaming, IoT, Real-time applications

3. Amazon Redshift

- Cloud-based data warehousing service.
- **Features** - Large-scale data analytics, High query performance
- **Used For** Business intelligence, Big data analytics

4. Amazon Aurora

- High-performance relational database compatible with MySQL and PostgreSQL.
- **Features** - Faster performance, Automatic replication, Fault tolerance

5. Amazon ElastiCache

- In-memory caching service.
- **Supports** Redis, Memcached

- **Advantages** - Faster data access, Reduced database load

6. Amazon Neptune

- Graph database service.
- **Used For** Social networks, Recommendation systems, Fraud detection

7. Amazon DocumentDB

- Managed document database service compatible with MongoDB.

8. Amazon Keyspaces

- Managed Apache Cassandra-compatible database service.

Diagram of AWS Database Services -

AWS Database Services

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| RDS | DynamoDB | Redshift | Aurora |
| ElastiCache | Neptune | DocumentDB | Keyspaces |
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Advantages of Amazon Database Services - Scalability, Automatic backup, High availability, Security, Managed maintenance

AWS database services provide flexible, scalable, and fully managed solutions for relational, NoSQL, caching, graph, and analytics workloads in cloud environments.

Q3b) Explain Google Cloud Application in Detail

Google Cloud Platform (GCP) provides cloud-based services and applications for computing, storage, networking, databases, AI, and analytics.

Google Cloud applications help businesses develop, deploy, and manage applications using Google's global infrastructure. **Major Google Cloud Applications and Services** -

1. Google Compute Engine

- Provides virtual machines in the cloud.
- **Features** - Scalable infrastructure, High-performance computing

2. Google App Engine

- Platform-as-a-Service (PaaS) for developing and hosting applications.
- **Features** - Automatic scaling, Serverless deployment

3. Google Cloud Storage

- Object storage service for storing files and backups.

- **Features** - Secure storage, High durability

4. Google Kubernetes Engine (GKE)

- Managed Kubernetes container service.
- **Features** - Container orchestration, Automatic deployment and scaling

5. BigQuery

- Serverless data warehouse service.
- **Used For** Big data analytics, Business intelligence

6. Google Cloud SQL

- Managed relational database service.
- **Supports** MySQL, PostgreSQL, SQL Server

7. Firebase

- Backend platform for mobile and web applications.
- **Features** - Authentication, Real-time database, Hosting

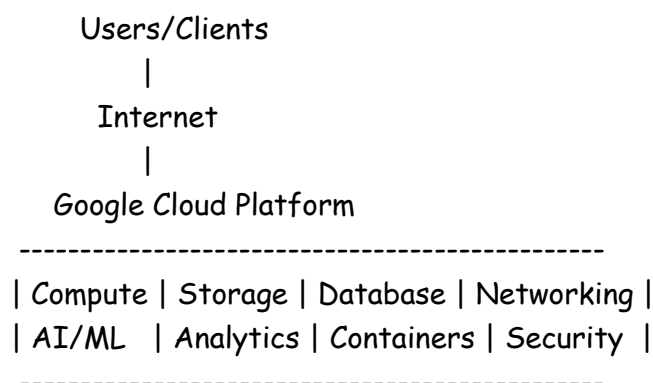
8. Google AI and Machine Learning Services

- **Services Include** - Vertex AI, Vision AI, Speech-to-Text, Translation API

9. Google Cloud Networking Services

- **Services Include** - Virtual Private Cloud (VPC), Load Balancer, Cloud CDN

Architecture of Google Cloud Applications -



Advantages of Google Cloud Applications - Scalability, High performance, Security, Global infrastructure, AI and analytics integration

Applications of Google Cloud - Web application hosting, Mobile app backend, Big data analytics, AI and ML applications, Cloud storage and backup

Google Cloud Platform provides powerful cloud applications and services for computing, storage, AI, analytics, and application development using Google's secure and scalable infrastructure.

Q4a) Identify the Different Components of Microsoft Azure and Explain Briefly

Microsoft Azure is a cloud computing platform provided by Microsoft that offers services for computing, storage, networking, databases, analytics, AI, and security.

Azure consists of several components that help organizations build, deploy, and manage applications through Microsoft-managed data centers.

Components of Microsoft Azure

1. Compute Component

- Provides processing power and virtual machines for running applications.
- **Services Include** - Azure Virtual Machines, Azure App Services, Azure Functions
- **Features** - Scalable computing, Supports Windows and Linux

2. Storage Component

- Provides cloud storage solutions for files, databases, backups, and applications.
- **Services Include** - Blob Storage, Disk Storage, File Storage
- **Features** - Secure storage, High availability

3. Networking Component

- Provides communication and connectivity services.
- **Services Include** - Virtual Network (VNet), Load Balancer, VPN Gateway
- **Features** - Secure communication, Traffic management

4. Database Component

- Provides managed database services.
- **Services Include** - Azure SQL Database, Cosmos DB, MySQL/PostgreSQL Services
- **Features** - Automatic backup, High scalability

5. Identity and Security Component

- Provides authentication, authorization, and security management.
- **Services Include** - Azure Active Directory (Azure AD), Security Center
- **Features** - Identity management, Access control

6. Analytics Component

- Used for data processing and business analytics.
- **Services Include** - Azure Synapse Analytics, HDInsight
- **Features** - Big data processing, Data visualization

7. AI and Machine Learning Component

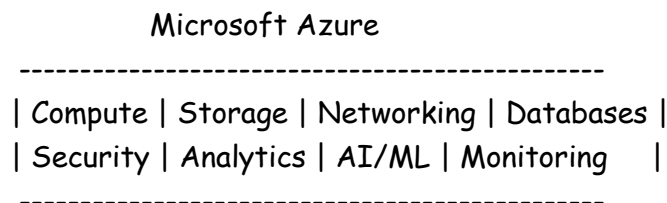
- Provides AI and machine learning tools.

- **Services Include** - Azure Machine Learning, Cognitive Services
- **Applications** - Speech recognition, Image processing

8. Management and Monitoring Component

- Used to monitor and manage Azure resources.
- **Services Include** - Azure Monitor, Azure Automation

Diagram of Microsoft Azure Components



Advantages of Azure Components - Scalability, Security, High availability, Easy integration, Cost effectiveness

Microsoft Azure provides various components for computing, storage, networking, databases, AI, and security, enabling organizations to build scalable and secure cloud applications.

b) How Cloud Computing Can Be Used for Business and Consumer Applications like ERP or CRM?

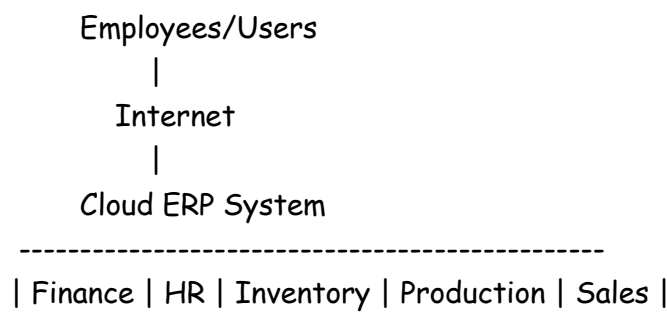
Cloud computing provides on-demand access to computing resources and applications over the internet. Businesses and consumers use cloud services for applications such as ERP and CRM without maintaining expensive infrastructure.

1. ERP (Enterprise Resource Planning)

ERP is software used to manage business processes such as: Finance, Human resources, Inventory, Production

Cloud-based ERP allows organizations to access ERP applications through the internet.

Cloud ERP Architecture



Advantages of Cloud ERP - Reduced infrastructure cost, Real-time data access, Scalability, Remote accessibility, Automatic updates

Examples - SAP Cloud ERP, Oracle ERP Cloud

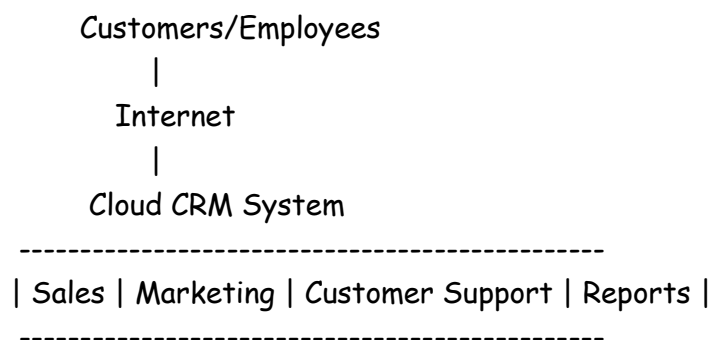
2. CRM (Customer Relationship Management)

CRM software manages customer interactions and business relationships.

Cloud CRM stores customer data on cloud servers and provides online access.

Functions of CRM - Customer management, Sales tracking, Marketing automation, Customer support

Cloud CRM Architecture



Advantages of Cloud CRM - Easy customer data access, Better customer service, Improved collaboration, Lower maintenance cost, Mobility support

Examples - Salesforce, Zoho CRM, Microsoft Dynamics CRM

Other Consumer Applications of Cloud Computing -

Application	Usage
Gmail	Email services
Google Drive	Cloud storage
Netflix	Video streaming
Online Gaming	Cloud gaming platforms
E-learning	Online education systems

Benefits of Cloud Computing for Business and Consumers -

- 1. Cost Reduction** - No need for expensive hardware.
- 2. Scalability** - Resources can be increased or decreased easily.
- 3. Accessibility** - Applications accessible from anywhere.

4. Automatic Updates - Cloud providers manage software updates.

5. Data Backup and Recovery - Improves data protection and disaster recovery.

Cloud computing enables businesses and consumers to use applications like ERP and CRM efficiently through scalable, cost-effective, and internet-based services with improved accessibility and reliability

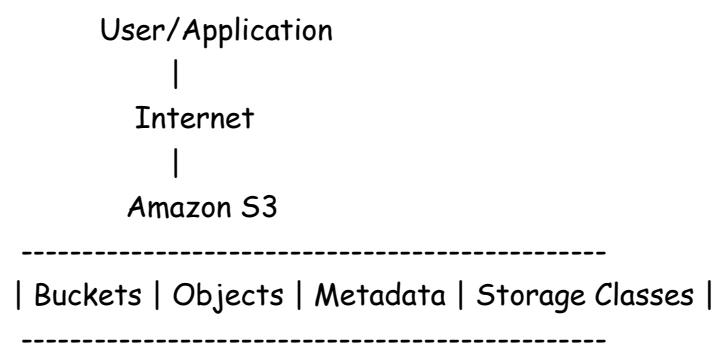
Q3b) Explain Amazon S3 and Amazon EC2

1. Amazon S3

Amazon S3 (Simple Storage Service) is an object storage service provided by AWS used to store and retrieve data through the internet.

It stores: Files, Images, Videos, Backups, Application data

Architecture of Amazon S3



Components of S3

1. Bucket - Logical container used to store objects.

2. Object - Actual file/data stored in bucket.

3. Metadata - Information about stored objects.

Features of Amazon S3 - High durability, Scalability, Secure storage, Backup and recovery, Lifecycle management

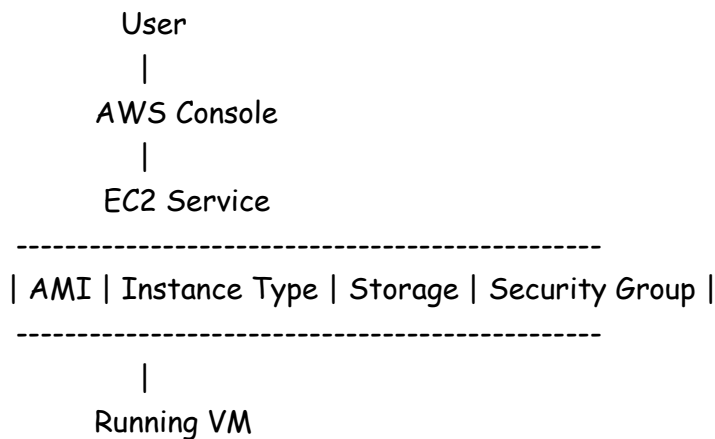
Applications of S3 - Cloud backup, Data archiving, Media hosting, Big data storage

Advantages of S3 - Pay-as-you-use, Easy accessibility, High availability, Automatic scaling

2. Amazon EC2

Amazon EC2 (Elastic Compute Cloud) provides virtual servers called instances for running applications in the cloud.

Architecture of EC2



Components of EC2

1. **AMI (Amazon Machine Image)** - Template containing OS and software.
2. **Instance** - Virtual machine running in cloud.
3. **Security Group** - Acts as firewall for EC2 instances.
4. **EBS Storage** - Persistent storage attached to EC2.

Features of EC2 - Scalable computing, Flexible instance types, On-demand resource allocation, Remote access

Applications of EC2 - Web hosting, Application deployment, Development/testing, Cloud servers

Advantages of EC2 - Cost effective, Easy deployment, High scalability, Reliable infrastructure

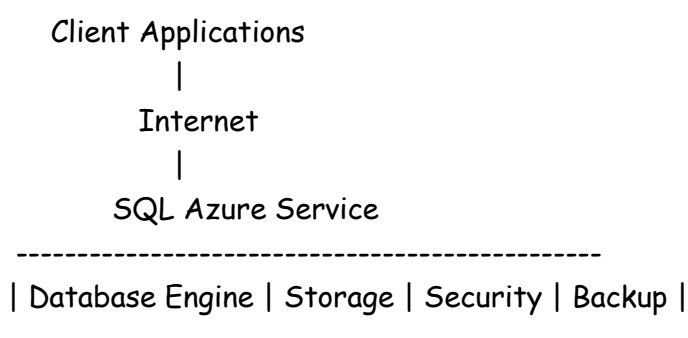
Amazon S3 provides scalable cloud storage while Amazon EC2 provides virtual computing resources for running cloud applications.

Q3c) Explain SQL Azure in Detail

Microsoft Azure SQL Azure, now called **Azure SQL Database**, is a cloud-based relational database service provided by Microsoft.

It is based on SQL Server technology and supports database management in cloud environments.

Architecture of SQL Azure



Features of SQL Azure -

1. **Fully Managed Database** - Microsoft handles: Installation, Maintenance, Updates, Patching
2. **Scalability** - Database resources can be scaled easily.
3. **High Availability** - Automatic replication and failover support.
4. **Security** - Data encryption, Firewall protection, Access control
5. **Automatic Backup** - Regular backups and recovery support.
6. **Multi-Tenant Architecture** - Multiple users share cloud infrastructure securely.

Components of SQL Azure

1. **Database** - Stores relational data in tables.
2. **SQL Engine** - Processes SQL queries.
3. **Data Center** - Stores data across distributed servers.
4. **Management Portal** - Used for database configuration and monitoring.

Advantages of SQL Azure - Reduced administration, Scalability, Reliability, Cost effectiveness, Remote accessibility

Applications - Enterprise applications, Web applications, ERP systems, CRM systems

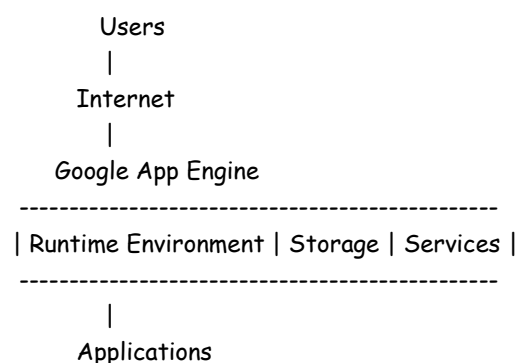
SQL Azure is a scalable, secure, and fully managed cloud database service that simplifies database management and supports modern cloud applications.

Q4a) Explain Google App Engine with its Installation Steps

Google App Engine (GAE) is a Platform as a Service (PaaS) provided by Google Cloud Platform for developing and hosting web applications.

It automatically manages infrastructure, scaling, and server maintenance.

Architecture of Google App Engine



Features of Google App Engine - Automatic scaling, Load balancing, Serverless deployment, High availability, Built-in security

Supported Languages - Python, Java, Node.js, PHP, Go

Services Provided by GAE - Datastore, Authentication, Memcache, Task queues

Installation Steps of Google App Engine -

Step 1: Create Google Cloud Account

- Sign in to Google Cloud Platform.

Step 2: Install Google Cloud SDK

- Download and install Google Cloud SDK.

Step 3: Configure SDK

Run command: `gcloud init`

Step 4: Create New Project

- Create project from Google Cloud Console.

Step 5: Enable App Engine

Run: `gcloud app create`

Step 6: Create Application Files

- Create app source code.
- Create app.yaml configuration file.

Step 7: Deploy Application

Run: `gcloud app deploy`

Step 8: Access Application

Open browser: `gcloud app browse`

Advantages of Google App Engine - No server management, Automatic scaling, Fast deployment, Cost efficient

Applications - Web applications, Mobile backends, APIs

Google App Engine simplifies application deployment and management using Google's scalable cloud infrastructure.

Q4c) Differentiate Between Amazon DynamoDB and Amazon S3

Parameter	DynamoDB	Amazon S3
Type	NoSQL Database Service	Object Storage Service
Data Structure	Key-value and document data	Files and objects
Usage	Stores structured data	Stores unstructured data
Access Method	Query-based access	Object/file access
Performance	Millisecond latency	Slower compared to DynamoDB
Scalability	Automatic scaling	Virtually unlimited storage
Data Model	Tables, items, attributes	Buckets and objects
Best For	Real-time applications	Backup and file storage
Query Support	Supports database queries	No advanced query support
Examples	Gaming, IoT, shopping carts	Images, videos, backups

Q4a) Enlist Applications of Cloud Computing in Different Areas. Describe Any Two Applications.

Cloud computing provides on-demand computing resources such as servers, storage, databases, networking, and software over the internet. It is widely used in different fields because of: Scalability, Cost efficiency, Remote accessibility, High availability

Applications of Cloud Computing in Different Areas -

Area	Application
Education	E-learning platforms
Healthcare	Online patient records
Banking	Online banking systems
Business	ERP and CRM applications
Entertainment	Video streaming services
Government	E-governance services
IT Industry	Cloud hosting and storage
Research	Scientific simulations
Social Media	Online communication platforms
E-commerce	Online shopping systems

1. Cloud Computing in Education - Cloud computing enables online learning and educational resource sharing through the internet.

Features - Online classes, Digital libraries, Assignment submission, Virtual labs

Examples - Google Classroom, Microsoft Teams, Coursera

Advantages - Learn from anywhere, Reduced infrastructure cost, Easy collaboration, Scalable learning platforms

2. Cloud Computing in Healthcare - Healthcare organizations use cloud computing to store and manage patient data and provide online medical services.

Features - Electronic Health Records (EHR), Telemedicine, Online appointment systems, Medical data storage

Advantages - Fast access to patient records, Improved collaboration between hospitals, Data backup and recovery, Reduced operational cost

Examples - Online consultation systems, Cloud hospital management systems

Cloud computing is widely used in education, healthcare, business, entertainment, banking, and many other areas to provide scalable, reliable, and cost-effective services.

b) Identify the Different Components of Amazon Web Services

AWS consists of multiple cloud service components that provide computing, storage, networking, databases, security, analytics, and management services.

These components help users build and manage cloud applications efficiently.

Components of AWS

1. Compute Component

Provides virtual computing resources.

Services Include - EC2, Lambda, Elastic Beanstalk

2. Storage Component

Provides cloud storage services.

Services Include - Amazon S3, EBS, Glacier

3. Database Component

Provides managed database services.

Services Include - RDS, DynamoDB, Redshift

4. Networking Component

Provides communication and networking services.

Services Include – VPC, Route 53, Elastic Load Balancer

5. Security Component

Provides authentication and security management.

Services Include – IAM, AWS Shield, KMS

6. Analytics Component

Used for big data processing and analytics.

Services Include – Athena, EMR, QuickSight

7. Management and Monitoring Component

Used for resource monitoring and automation.

Services Include – CloudWatch, CloudFormation

8. Application Integration Component

Supports communication between applications.

Services Include – SNS, SQS

9. Machine Learning and AI Component

Provides AI and machine learning services.

Services Include – SageMaker, Rekognition

Diagram of AWS Components

AWS Components

| Compute | Storage | Database | Networking | Security |
| Analytics | Monitoring | AI/ML | Integration |

Advantages of AWS Components – Scalability, Flexibility, Reliability, High security, Pay-as-you-use model

AWS provides a wide range of components and services for computing, storage, networking, databases, security, analytics, and AI to build scalable cloud-based applications efficiently.